Education

University of Georgia — Georgia, US
Ph.D. in Computer Science — *Sep 2025 – Jun 2030*
Core courses: Software engineering, Operating system, and Programming languages

Fudan University — Shanghai, China
M.Sc. in Aerospace Science and Technology; ***GPA: 3.41/4.00 (Ranked*** 1th ***in 45)*** — *Sep 2022 – Jun 2025*
Core courses: Self-repairing flight control system, Atmospheric flight dynamics, and Aircraft system identification

Shenyang Aerospace University — Shenyang, China
B.E. in Computer Science and Technology; ***GPA: 8.28/10 (92/100, Ranked*** 1th ***in 155)*** — *Sep 2018 – Jun 2022*
Core courses: Principles of Computer Organization, Operating Systems, and Data structures and algorithms

Research projects & Interests

My research focuses on improving the interpretability, controllability, and trustworthiness of AI systems across foundation models and safety-critical applications.

1. Synthesizing Diverse Data in Feature Space of LLMs

- Data diversity is crucial for effective LLM post-training, but common metrics (e.g. Distinct-n, Cosine Distance) often fail to reflect downstream task performance. We proposed Feature Activation Coverage (FAC) as a superior diversity metric that better predicts task performance by measuring coverage of task-relevant features.
- Guided by FAC, we synthesize a small amount of targeted data to fill coverage gaps, achieving superior performance on Toxicity Detection, Reward Modeling, Behavior Steering, and Instruction Following. Notably, on Instruction Following, our method matches prior SOTA MAGPIE while using 150× less data. Project page: `Less is Enough`.

2. Interpretable Fault Detection in Safety-Critical Time-Series Systems

- Achieved 98.99% accuracy in aircraft sensor fault detection with compact (4.58 MB) explainable models via knowledge transfer and pruning. Published in *IEEE Transactions on Aerospace and Electronic Systems*.
- Proposed Neural Processes-based Deep Transfer Learning with Graph Convolution Network for intelligent fault detection, which leverages Neural Processes' few-shot adaptation (joint global/task-shared and local/per-sample latent modeling) to minimize target-domain data demand, provide multi-scale uncertainty estimates (task-wide and per-sample). Published in *Neurocomputing*.

Intern Experience

Huawei Technologies Co., Ltd — Shanghai, China
AI Engineer Internship — *Jul 2023 – Aug 2023*

- Visualize changes in the number of cores and proportions of different types of virtual machines over time and analyze recent market demand changes.
- VM virtual machine request sequence prediction: Propose an XAI prediction ecological model of multimodal information integration, use machine learning algorithms and an improved Transformers algorithm that can encode different variables (static variables, observed historical time-varying variables and prior time-varying variables) and has interpretability to complete the VM request sequence Forecast, and complete the validity test of the algorithm on different time scales (monthly level, yearly level), data with emergencies and business information.

Guoxuan High-Tech Co., Ltd (TOP3 for Power battery) — Shanghai, China
Algorithm Internship — *Sep 2022 – Jun 2023*

- Developed battery fault detection systems utilizing both supervised and unsupervised learning approaches for LiFePO4 and ternary lithium batteries, with specialized focus on thermal runaway early warning mechanisms.
- Innovated an enhanced thermal runaway detection algorithm using Stacking ensemble method:
  (1) Validated algorithm efficacy through rigorous testing;
  (2) Filed patent application for the novel methodology;
  (3) Implemented cloud-based deployment via Alibaba Cloud PAI platform.
- Created remaining useful life (RUL) prediction models employing LSTM neural networks and other intelligent algorithms for battery lifespan forecasting.

Dongguan Securities Co., Ltd Shanghai, China

Quantitative Intern *May 2022 – Aug 2022*

- Engineered high-performance data transfer solutions for large-scale financial datasets (30 GB+), overcoming FTP bandwidth limitations through protocol optimization and implementing automated checksum validation to ensure data integrity.
- Developed and implemented fundamental quantitative strategies including dual moving average crossover, multi-factor stock selection (PEG ratio, valuation metrics), and Bollinger Band trading approaches. Performed comprehensive data preprocessing and statistical analysis on market data using Python and C++.

Selected Publication & Academic Services

[1]Li Z, Wu X, Li Y, Hu L, et al. Less is Enough: Synthesizing Diverse Data in Feature Space of LLMs[J]. arXiv, 2026. Submitted to a conference (under review).

[2]Li Z, Tu J, Zhu J, et al. Scalable and reliable deep transfer learning for intelligent fault detection via multi-scale neural processes embedded with prior knowledge[J]. Neurocomputing, 2025: 131364.

[3]Li Z, Fan R, Ma J, et al. Dynamic Temporal Denoise Neural Network with Multi-Head Attention for Fault Diagnosis Under Noise Background[J]. Sensors, 2024, 24(21): 6813.

[4]Li Z, Zhang Y, Ai J, et al. A lightweight and explainable data-driven scheme for fault detection of aerospace sensors[J]. IEEE Transactions on Aerospace and Electronic Systems, 2023, 59(6): 8392-8410.

[5]Qu N, Li Z, Li X, et al. Multi-parameter fire detection method based on feature depth extraction and stacking ensemble learning model[J]. Fire Safety Journal, 2022, 128: 103541.

[6]Li Z, Huang Y, Li X, et al. Wildland fire burned areas prediction using long short-term memory neural network with attention mechanism[J]. Fire technology, 2021, 57: 1-23.

Reviewer for *IEEE Transactions on Artificial Intelligence*, *IEEE Transactions on Industrial Informatics*, *IEEE Internet of Things Journal*, and various other journals and conferences including AAAI.

Awards & Achievements

Postgraduate period Shanghai, China

*Received **6** scholarships* *Sep 2022 – Jun 2025*

- Qingyun Scholarship (*Awarded annually to only 10 graduate students*) Oct 2024
- National Scholarship *(Only 1 scholarship recipient is selected from 45 students)* Jun 2023
- Yang Yongming Scholarship Dec 2023
- Inspirational Scholarship of Zhang Mingwei Dec 2022

Undergraduate period Shenyang, China

*Received **12** scholarships and **10** national level competition awards* *Sep 2018 – Jun 2022*

- National Scholarship (*Only 1 recipient is selected from 155 students*) Dec 2020
- Hexagon Enterprise Scholarship Dec 2021
- Academic Scholarship of First Class Jun 2021
- Academic Scholarship of First Class Dec 2021
- School Annual Personality Scholarship (*Awarded annually to only 10 undergraduate students*) Dec 2021
- National Physics Academic Competition **Second Prize** Aug 2019
- National Student Mathematics Competition **Second Prize** Nov 2019
- National Students Asia Pacific Mathematical Modeling Competition **Second Prize** Dec 2019
- National Students' May Day Mathematical Modeling Competition **First Prize** Jun 2020
- National Student Mathorcup Mathematical Modeling Competition **First Prize** ($\approx 5\%$) Jun 2020

Skills

Languages

- English: TOEFL, CET-6

Technology

- Python (PyTorch), SQL, LaTeX, MATLAB, MS Office. Experience with LLM fine-tuning and evaluation, including LoRA, custom training objectives, and data preprocessing pipelines.